NCI
Coláiste Náisiúnta na hÉireann | National College of Ireland

This certificate is awarded to:

Yaroslav Kondaurov

FOR COMPLETING THE PROGRAMME

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Data Structures CA1
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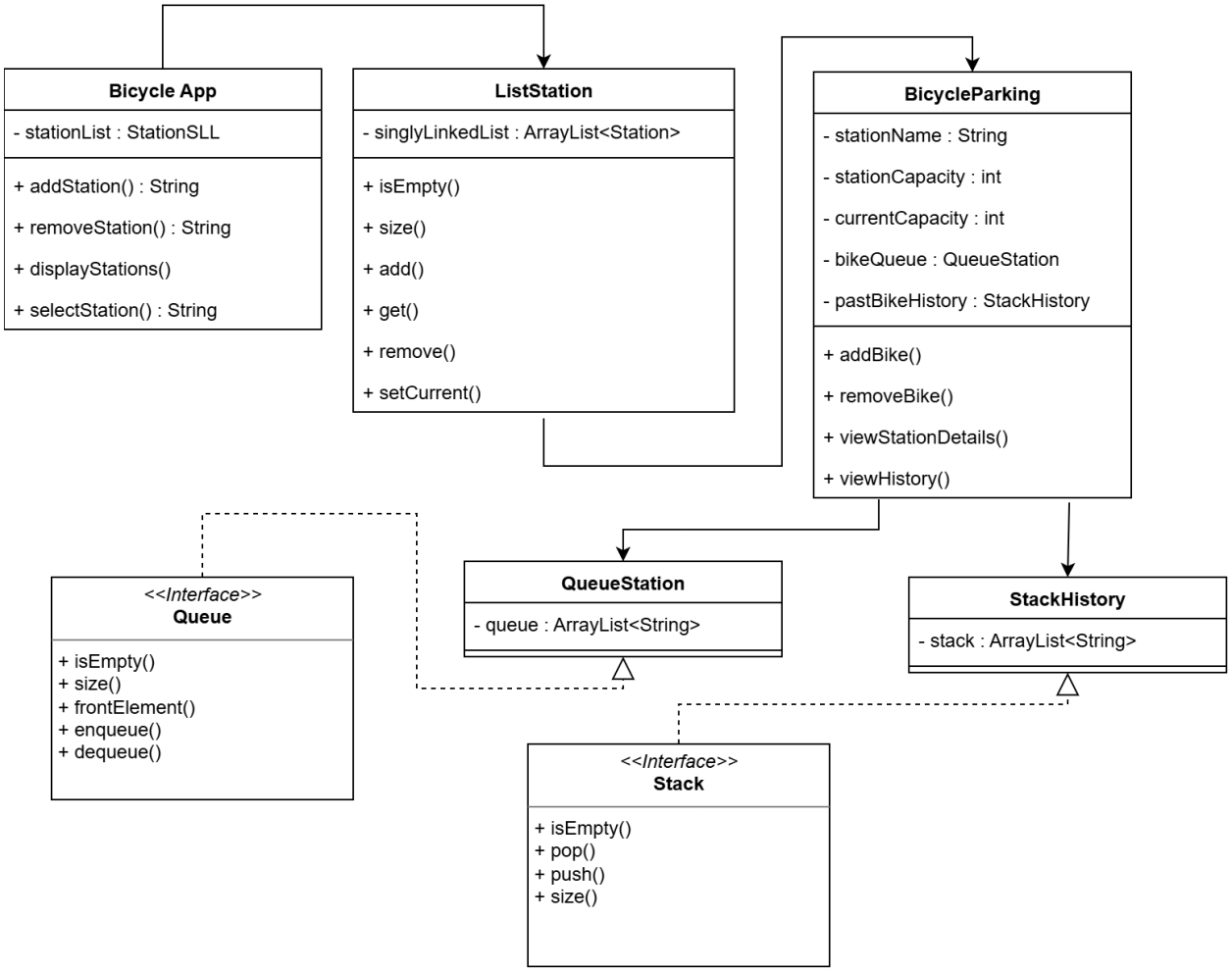
Application Concept:
The theme chosen for this project is all about Mobility Innovation from the Smart Docklands Pilot Call themes. Overall, this application's theme revolves around expanding cycling options for people going in and out of the docklands or staying within by providing said people with an app to use.

This application is primarily designed for cyclists to use to help in their cycling needs. The core features of the application that will help in this endeavor is the ability to track all bicycle parking stations throughout the docklands region. It will achieve this by storing all the parking spots through a singly linked list. The user can add or remove parking stations from a singly linked list as well as show a display of all stations. This allows cyclists to save the stations they use the most and remove unused stations. The users can then select a station and get presented with additional options.

The second feature is the ability to reserve a bicycle spot at any of one of the parking stations throughout the dockland area if the parking station is at full capacity. To facilitate this role, the application will use a priority queue system. If there is no queue at a specific station, the cyclist is notified of a free space and the queue is freed up.

The third feature is showing a history of all bikes that were parked at a specific station. When a bike leaves a station, it is stored onto a stack as history where an admin can view which stations have had the most bikes leave and therefore find the most used stations within the dockland.

Class Diagram:



Github Repo Link:
https://github.com/Arleits/Data-Structures-Algorithms-CA1